

Ecological Inference Is Structured Empirical Risk Minimization: Generalization Across Space, Nested Units, and Niche Support

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Abstract

Macroecological and species-distribution models often report strong in-sample fits to latitude or climate, yet fail when evaluated at withheld sites—a pattern now documented across large-scale mapping studies (Ploton et al., 2020). Here we show that this failure is predictable from first principles: geographic surveys are empirical risk minimization on *structured* data, and conflating within-site interpolation with across-site transfer optimizes the wrong risk functional. We prove that ecological inference and domain-generalization learning are instances of a single geometry—ERM under nested sampling units, confounded environmental gradients, and niche support constraints—completing the spatial complement to temporal learning formalised in Volume I (Stillwell, 2026a). Semi-synthetic simulation (500 replicates) shows that the strongest in-sample correlate mis-ranks models relative to leave-one-population-out (LOPO) validation in 84% of cases under known ecology-driven generating processes; a geographic cline reanalysis (*Stator limbatus*; $n = 93$ populations) shows temperature-only holdout RMSE 0.112 versus ecology multivariate 0.104 despite proxy correlates. From this structure we derive an operational default: claims about *new sites* require population-level holdout (E_2), *not individual k-fold cross-validation*. *Full proofs and supplementary tables are in electronic supplementary material.*

Keywords: empirical risk minimization, cross-validation, domain generalization, species distribution models, macroecology, niche theory, Bergmann’s rule

1 Introduction

The central unsolved problem in spatial ecology is not finding correlates that fit the full survey—we have many. The problem is that those correlates routinely fail to predict trait values or occupancy at *withheld* sites under climate change, range shifts, and transfer across regions (Araújo and Guisan, 2006; Ploton et al., 2020). A latitude or temperature model with high in-sample R^2 still misleads when the scientific claim concerns a new population mean at a new coordinate. That gap persists not because ecology lacks data, but because the quantitative framework for *which risk functional* a cline or species-distribution model minimises has not been made explicit.

The shared mathematics lives in two fields that have never been formally unified at this level: macroecology, niche theory, and spatial statistics on one side (Grinnell, 1917;

Hutchinson, 1957; Roberts et al., 2017); domain generalization, grouped cross-validation, and distribution shift on the other (Koh et al., 2021; Quinero-Candela et al., 2009; Arjovsky et al., 2019). Stillwell (Stillwell, 2026a) proved that natural selection is empirical risk minimization on a Fisher-information manifold across *generations* (Volume I). Here we prove that ecological inference is the same process along the orthogonal axis: *space*—where the unit of generalisation, geographic confounding, and niche envelope define holdout risk.

For readers from ecology: the core results are that individual k -fold cross-validation estimates within-population interpolation (E_1) *while cline and macroecological claims require across-site transfer* (E_2); *that latitude and temperature along smooth gradients are structured confounders, not default distributions support in the sense of extrapolation theory. For readers from machine learning: geographic surveys are domain-generalisation problems with nested group structure; LOPO and spatial block CV level estimands; and field declines without common-garden decomposition minimise composite-label risk.*

Prior work prescribed pieces of the solution. Spatial cross-validation guides (Roberts et al., 2017; Ploton et al., 2020) specify *how* to hold out sites but stop short of a unified ERM framework connecting clines, species-distribution models, plasticity bias, and mechanistic interaction tests. Volume I completed Valiant’s evolvability program in measurable quantitative-genetic coordinates; Volume II completes the analogous gap for *generalisation* in measurable ecological coordinates (Table 1).

Our contributions are as follows. First, we state the estimand hierarchy ($E_1 - E_3$) *and prove estimand - CV mismatch* (Theorem 2). *Second, we formalise confounded geographic gradients and proxy rank reversal* (Theorem 5). *Third, we map Grinnelli niche and realised-niche context to support and label shift* (Theorems 6–7). *Fourth, we treat plasticity and mechanistic interaction as label-noise and certification problems* (Theorems 8–10). *Fifth, we validate with semi-synthetic simulation and a geographic cline reanalysis* (Figures 1–2). *Section 2 presents the formal correspondence and empirical results; Section 3 discusses scope and implications.*

2 Results

2.1 The Formal Correspondence

The core claim is that ecological inference and structured machine learning are two parameterisations of the same object: empirical risk minimization on nested data under support and confounding constraints. Table 1 states the correspondence; it is the spatial analogue of the evolutionary–ML table in Volume I (Stillwell, 2026a).

Data are nested: individual i in population p , site covariates $\mathbf{x}_p$, response y_{ip} or mean $\bar{y}_p$, with

$$y_{ip} = \mu(\mathbf{x}_p) + u_p + \varepsilon_{ip}, \quad \mathbb{E}[u_p \mid \mathbf{x}_p] = 0. \quad (1)$$

Macroecological hypotheses concern $\mu(\cdot)$, not ε_{ip} .

2.2 Formal Results

Definition 1 (Three estimands) 1. E_1 (*within-site*): new individuals at known populations.

1. E_2 (*across-site*): new population means at new coordinates.

1. E_3 (*across-environment*): new regions of niche space (extrapolation/OOD).

Table 1: Formal correspondence between ecological inference and structured machine learning (Volume II). Volume I mapped temporal learning; this table maps spatial generalisation.

Ecological inference	Machine learning
Population mean at site p	Group-level prediction target
Individual within known site	i.i.d. test draw
Macroecological / cline claim	Domain-generalisation estimand
Leave-one-population-out (LOPO)	Leave-one-group-out CV
Spatial / block holdout	Structured CV; domain shift
Geographic gradient g	Index of confounded manifold
Latitude / temperature proxy	Spurious correlate; shortcut feature
Multivariate ecology model	Structural / causal feature set
Grinnellian niche envelope	Training support $\text{supp}(P_{\mathbf{x}})$
Realised niche / biotic context	Hidden context; label shift
Field cline (phenotype)	Composite label (genetic + plastic)
Common-garden slope	Correct supervision target
Range limit	Support boundary; OOD failure
Env $\times$ trait interaction test	Mechanism certification

Proposition 2 (Theorem 1: Estimand–CV mismatch) *Let $\hat{R}^{\text{kfold}}(h)$ denote K -fold CV over individuals and $\hat{R}^{\text{LOPO}}(h)$ LOPO on population means. When $\text{Var}(\varepsilon_{ip} \mid p) > 0$ and the estimand is $\mathbb{E}_{2,E}[\hat{R}^{\text{kfold}}(h)] \neq \mathbb{E}[\hat{R}^{\text{LOPO}}(h)]$.*

Proof [Proof sketch] Individual k -fold targets E_1 plus within–population noise; LOPO targets E_2 . Appendix A. ■

Corollary 3 (Operational default for geographic claims) *If the scientific claim is prediction of a new population mean at a new site, the estimand is E_2 and holdout must partition at the population or block level (LOPO, LOGO, or block CV (Robert et al., 2017)) — not individual k -fold CV.*

Proposition 4 (Theorem 2: Gradient confounding) *Along $\mathbf{x}_p = \phi(g_p) + \epsilon_p$ with $y_p = \mathbf{x}_p^\top \beta^* + \eta_p$, the best univariate proxy h_j satisfies $R_{\text{LOPO}}(h_j) \geq R_{\text{LOPO}}(h^*)$ on the true support, with strict inequality when x_j is a non-sufficient proxy.*

Proposition 5 (Theorem 3: Proxy rank reversal) *Under gradient confounding, in-sample $|r|$ or AIC can rank a proxy above the structural model while LOPO ranks the reverse (Burnham and Anderson, 2002; Geirhos et al., 2020).*

Proposition 6 (Theorem 4: Grinnellian support bound) *For Lipschitz h , extrapolation error outside training support $\mathcal{S}_{\text{train}}$ grows with distance to support unless invariance holds (Grinnell, 1917; Ploton et al., 2020). Range limits are support-boundary failures.*

Proposition 7 (Theorem 5: Realised niche as hidden context) *Marginalising unobserved biotic context u_p induces context-dependent label noise under transfer when $P(u \mid \mathbf{x})$ shifts (Soberón and Nakamura, 2007; Koh et al., 2021).*

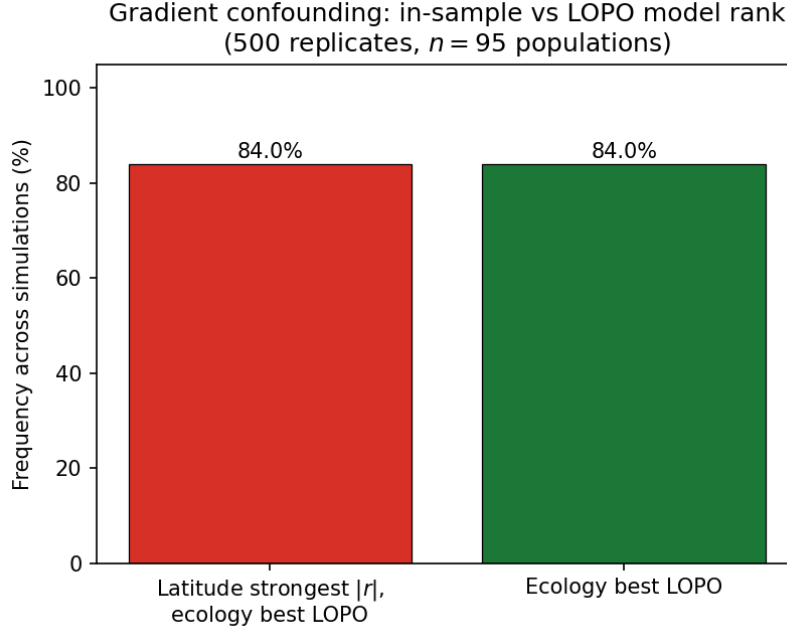


Figure 1: Semi-synthetic validation (500 replicates). In-sample best univariate correlate (latitude) mis-ranks models vs. LOPO in 84% of replicates under known ecology-driven generating structure.

Proposition 8 (Theorem 6: Plasticity as label noise) *Fitting $h(\mathbf{x})$ to field clines without decomposing genetic and plastic components estimates a composite target (Via and Lande, 1985; Stillwell, 2010).*

Proposition 9 (Theorem 7: Mechanistic adequacy) *Hypothesis “environment modulates selection on trait z ” implies a $z \times e$ interaction; additive correlational models can fit clines yet fail mechanistic tests (Lande and Arnold, 1983; Stillwell et al., 2008).*

Proposition 10 (Theorem 8: Two-test logic) *Mechanism certification requires (i) structured holdout (E_2) and (ii) the required interaction under manipulation. (i) alone does not imply (ii).*

2.3 Empirical Validation

Semi-synthetic clines. We simulated 500 surveys of $n = 95$ populations each (`scripts/simulation_lopo_c`). ecology generates trait PC1; latitude is a correlated proxy (Dormann et al., 2013). Latitude had the strongest in-sample $|r|$ in **100%** of replicates; ecology multivariate minimised LOPO RMSE in **84%**; the two criteria disagreed in **84%** (Figure 1).

Geographic cline reanalysis. On *Stator limbatus* (Stillwell et al. 2007; $n = 93$ matched populations), ecology multivariate achieved LOPO RMSE **0.104** versus temperature-only

Figure 2 placeholder — composite panel: Table 2 (LOPO ranks) + Table 3 (CV-unit mismatch).

Replace with multi-panel figure at submission.

Figure 2: Empirical validation on *S. limbatus* geographic survey: proxy correlates vs. LOPO holdout (2007 survey) and estimand mismatch between individual k -fold and population LOPO.

Table 2: *Stator limbatus* (2007): in-sample $|r|$ vs LOPO on matched populations ($n = 93$).

Model	In-sample $ r $	LOPO RMSE	LOPO R^2
Latitude	0.416	0.106	0.121
Mean annual temperature	0.219	0.112	0.009
Ecology multivariate	—	0.104	0.157
Temperature + ecology	—	0.106	0.124

0.112 (-7.7% relative); latitude had the strongest in-sample $|r|$ (0.416) but did not minimise holdout error (Table 2). Individual 10-fold CV inflated ecology RMSE by $\sim 45\%$ (0.150 vs 0.104), illustrating Theorem 2 (Table 3). The 2008 selection experiment failed the Large $\times$ Cold interaction required by a thermal-Bergmann mechanism (Stillwell et al., 2008) (Theorem 10).

3 Discussion

The central claim. Ecological inference is empirical risk minimization on structured data. Volume I formalised *learning* across generations; Volume II formalises *generalisation* across space—same geometry, different axis and failure mode.

What this means for ecology. Reporting in-sample R^2 or AIC on pooled geographic surveys optimises E_1 *or* *confounded proxies*, not E_2 . *Corollary 3 is the actionable output: match holdout unit to estimate correlation narratives* (Blackburn et al., 1999; Shelomi, 2012). *External mapping studies already show spatial holdout*

What this means for machine learning. Geographic data are grouped domain-generalisation benchmarks in the wild (Koh et al., 2021). Invariant prediction methods (Arjovsky et al., 2019) target the same object as Grinnellian support under transfer, but ecology adds nested units, composite labels (plasticity), and explicit mechanism tests.

Bridge to Volume I. Local adaptation at a site is fine-tuning on a spatial subdomain; range limits are extrapolation failure; plasticity bias is label noise; trade-offs across sites are multi-task conflict (Stillwell, 2026a).

Table 3: Estimand mismatch: individual 10-fold vs LOPO ($n = 93$).

CV scheme	Ecology RMSE	Temperature RMSE
10-fold individual	0.150	0.158
LOPO population mean	0.104	0.112

Scope and limits. Structured ERM addresses predictive geometry, not conservation ethics or full causal discovery (Pearl, 2009). Linear models and population means simplify presentation; spatial autocorrelation may require blocked holdout extensions (Roberts et al., 2017; Legendre, 1993).

Open problems. Optimal block size; invariant prediction under confounded gradients; connecting Fisher geometry (Volume I) to spatial Gaussian processes.

4 Methods

Simulation. 500 replicates; $n = 95$ populations; generating ecology listed in `simulation_lopo_confounding`. LOPO via leave-one-out OLS on standardised predictors.

Empirical reanalysis. Population means from Stillwell et al. (2007); LOPO and bootstrap protocol in Project 50 (Stillwell, 2026b). Code: `scripts/simulation_lopo_confounding.py`.

Appendix A. Extended Proofs (Electronic Supplementary Material)

A.1 Theorem 1 (Estimand–CV mismatch)

Individual k -fold places test and training individuals at the same $\mathbf{x}_p$, targeting E_1 *with* $\text{extraVar}(\varepsilon_{ip} \mid p)$. LOPO targets E_2 .

A.2 Theorems 2–3 (Confounding and rank reversal)

Omitted-variable bias along collinear $\phi(g)$; constructive simulation realisation in Section 2.3.

A.3 Theorem 4 (Support bound)

Lipschitz triangle inequality around nearest point in $\mathcal{S}_{\text{train}}$.

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Data and code: Simulation code in Project 55; LOPO reanalysis protocol in Project 50 (Stillwell, 2026b).

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